

200 DAYS JAVA DSA SERIES

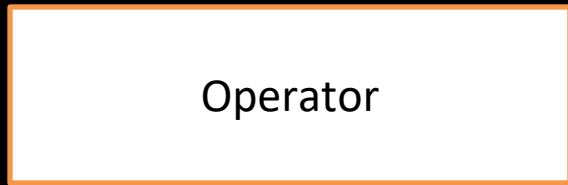
DAY 07: ASSIGNMENT OPERATORS SWAP TWO NUMBERS



Operator

10+20=30

operands



❑ Operators:

Operators are symbols that tell the compiler to perform specific operations, such as mathematical, logical, or bitwise operations, on data values. They act on variables or values to produce a result.

Example: In the expression $a + b$, the $+$ symbol is an operator that adds the values of a and b .

❑ Operands:

Operands are the values or variables on which the operator acts. In an expression, operands are the data items that operators manipulate.

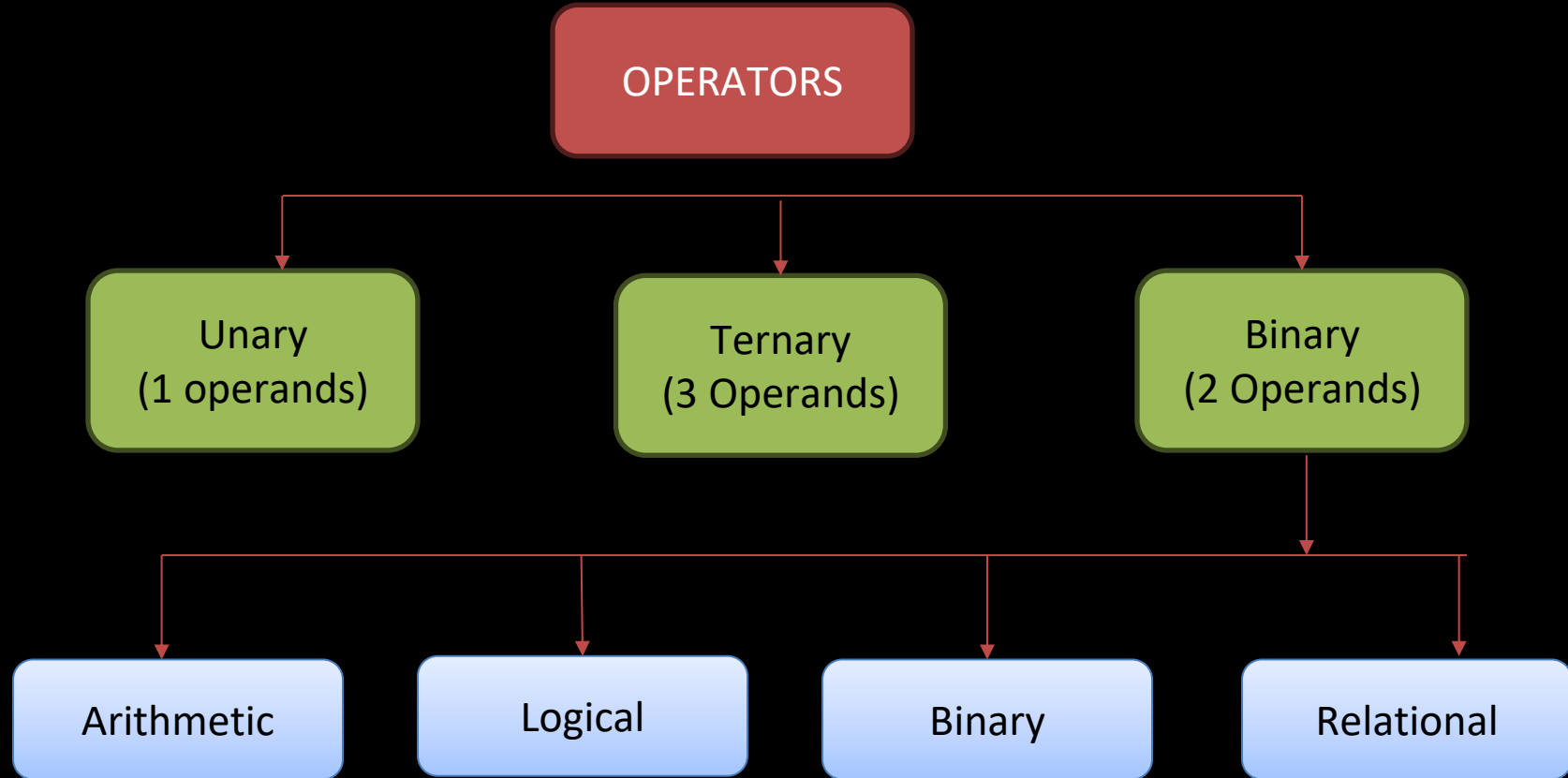
Example: In the expression $a + b$, a and b are operands, while $+$ is the operator acting on these operands.

□ Expression Example with Operators and Operands:

Consider $c = a + b * 2;$

- Operators: $=$, $+$, $*$
- Operands: a , b , 2 , and c

❑ Types Of Operators:



Unary
operator

Binary
Operator

Ternary
Operator

Operators	Type
++,--	Unary Operator
+, -, *, /, %	Arithmetic Operator
<, <=, >, >=, ==, !=	Relational operator
&&, , !	Logical Operator
&, , <<, >>, -, ^	Bitwise Operator
=, +=, -=, *=, %=	Assignment Operator
?:	

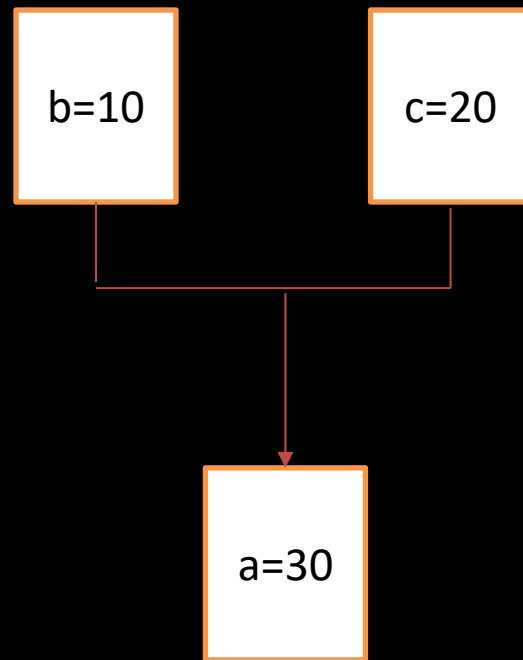
Assignment Operator(=)

20

~~10~~

a=b+c; valid

b+c=a; invalid



10=a



10=20



Lvalue=Rvalue

◆ Assignment Operator in Java:

Assignment operator ka use **variable me value assign karne ke liye** hota hai.

► Basic Syntax:

```
int a = 10; // Here, '=' is the assignment operator
```

LHS

Variable
e
a=10
(valid)

Literal
10=a
(invalid))

Expression
a+b=10(invalid

a= ; (RHS missing before ;)
=10;(Missing variable (LHS))
a=10(;is Missing)

◆ Types of Assignment Operators in Java:

Operator	Meaning	Equivalent Expression
=	Assign value	a = b
+=	Add and assign	a = a + b
-=	Subtract and assign	a = a - b
*=	Multiply and assign	a = a * b
/=	Divide and assign	a = a / b
%=	Modulus and assign	a = a % b

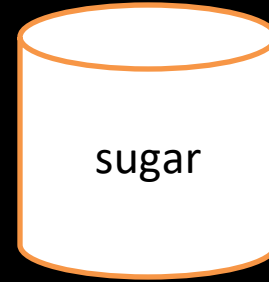
Java Programs on Assignment Operators:

- 1 Simple Assignment Operator (=)
- 2 Addition Assignment (+=)
- 3 Subtraction Assignment (-=)
- 4 Multiplication Assignment (*=)
- 5 Division Assignment (/=)
- 6 Modulus Assignment (%=)

"Swap of Two Numbers using Assignment Operators"

❖ **Method 1: Using Third Variable**

❑ Concept of Swapping



Swapping of number with temp variable

➤ Swapping with a Temporary Variable

1. Start
2. Input two numbers, a and b
3. Store a in temp
4. Set a = b
5. Set b = temp
6. Output the swapped values of a and b
7. End

**Before
swapping**

a=10

b=20

**After
swapping**

a=20

b=10

```
public class SwapWithTemp {  
    public static void main(String[] args) {  
        int a = 5, b = 10;  
  
        System.out.println("Before Swap: a = " + a + ", b = " + b);  
  
        int temp = a;  
        a = b;  
        b = temp;  
  
        System.out.println("After Swap: a = " + a + ", b = " + b);  
    }  
}
```

Method 2: Without Third Variable (Using + and - Assignment Operators)

Swapping of number without temp variable

➤ Swapping without a Temporary Variable

- Start
- Input two numbers, a and b
- Set $a = a + b$
- Set $b = a - b$
- Set $a = a - b$
- Output the swapped values of a and b
- End

**Before
swapping**

a=10

b=20

**Afther
swapping**

a=20

b=10

```
public class SwapWithoutTempAddSub {  
    public static void main(String[] args) {  
        int a = 5, b = 10;  
  
        System.out.println("Before Swap: a = " + a + ", b = " + b);  
  
        a += b; // a = a + b => a = 15  
        b = a - b; // b = 15 - 10 = 5  
        a -= b; // a = 15 - 5 = 10  
  
        System.out.println("After Swap: a = " + a + ", b = " + b);  
    }  
}
```

Method 3: Using XOR (Bitwise Assignment Operator)

```
public class SwapUsingXOR {  
    public static void main(String[] args) {  
        int a = 5, b = 10;  
  
        System.out.println("Before Swap: a = " + a + ", b = " + b);  
  
        a ^= b; // a = a ^ b  
        b ^= a; // b = b ^ a  
        a ^= b; // a = a ^ b  
  
        System.out.println("After Swap: a = " + a + ", b = " + b);  
    }  
}
```

jayesh_kande_ ▾ ●

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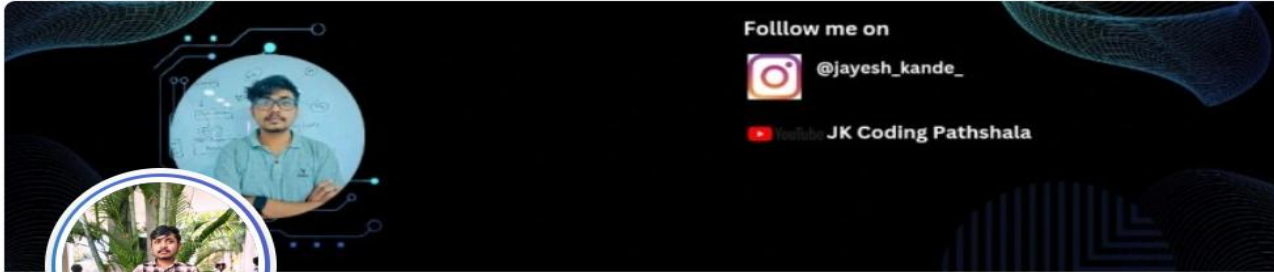
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Jayesh Kande

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